

THA WANG PHA, Thailand

WMO: 483150

Lat: 19.1231N

Lon: 100.8131E

Elev: 237

StdP: 98.52

Time Zone: 7.00 (E07)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.8	(c) 13.3	(d) 10.2	(e) 8.0	(f) 15.2	(g) 11.7	(h) 8.8	(i) 16.1	(j) 4.3	(k) 26.9	(l) 3.4	(m) 27.2	(n) 0.0	(o) N/A	(p) 0.279

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 12.9	(c) 37.7	(d) 25.7	(e) 36.5	(f) 25.7	(g) 35.4	(h) 25.6	(i) 28.2	(j) 33.8	(k) 27.7	(l) 33.1	(m) 27.3	(n) 32.4	(o) 1.0	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 26.7	(b) 23.0	(c) 31.4	(d) 26.3	(e) 22.4	(f) 30.7	(g) 26.0	(h) 22.0	(i) 30.2	(j) 92.5	(k) 34.0	(l) 90.2	(m) 33.3	(n) 88.3	(o) 32.7	(p) 30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3.5	3.2	2.8	9.2	39.5	1.9	1.4	7.8	40.6	6.7	41.4	5.7	42.2	4.3	43.2
			8.8	29.1	1.8	0.8	7.5	29.6	6.4	30.1	5.4	30.6	4.1	31.2
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.5	21.8	24.2	26.8	29.3	29.1	28.8	28.0	27.7	27.9	27.1	24.9	22.1
	DBStd	3.06	2.08	1.83	2.10	1.78	1.81	1.46	1.44	1.19	1.15	1.32	1.98	2.35
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	6	3	0	1	0	0	0	0	0	0	0	0	2
	CDD10.0	6016	366	397	522	579	593	564	557	548	537	531	447	376
	CDD18.3	2980	110	163	264	329	335	314	299	290	287	273	197	120
	CDH23.3	29948	864	1576	2852	3872	3766	3529	3074	2809	2776	2394	1554	882
(13)	CDH26.7	12659	252	746	1583	2176	1801	1455	1068	876	972	907	578	246
(14)	Wind	WSAvg	0.3	0.2	0.3	0.4	0.4	0.5	0.4	0.3	0.2	0.2	0.2	0.2
(15) Precipitation	PrecAvg	1501	14	11	45	102	179	178	285	325	235	78	27	15
	PrecMax	2007	67	62	150	187	374	392	499	553	402	167	147	89
	PrecMin	1207	0	0	0	21	59	75	173	163	139	17	0	0
	PrecStd	227	19	16	38	43	65	72	74	101	70	35	34	25
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.7	35.4	38.0	39.9	39.1	36.5	35.6	33.8	34.1	34.1	33.2	31.7
		MCWB	22.3	22.8	24.3	25.9	27.1	27.4	27.1	27.2	27.2	26.3	24.9	23.5
	2%	DB	30.5	34.0	36.6	38.3	36.8	35.0	33.7	32.6	33.1	32.9	32.1	30.3
		MCWB	21.9	22.8	24.0	25.7	27.0	27.1	27.0	26.9	26.8	25.9	24.5	22.9
	5%	DB	29.4	32.8	35.6	37.0	35.3	33.7	32.4	31.7	32.2	32.1	31.2	29.2
		MCWB	21.6	22.6	23.8	25.9	26.8	26.9	26.7	26.7	26.5	25.6	24.2	22.3
	10%	DB	28.2	31.5	34.3	35.6	33.8	32.4	31.2	30.7	31.3	31.1	30.2	28.1
		MCWB	21.2	22.5	23.8	25.9	26.6	26.6	26.4	26.4	26.3	25.3	23.7	21.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.3	26.1	27.3	28.6	28.8	28.5	28.3	28.0	28.1	27.5	26.2	25.1
		MCDB	28.5	32.2	34.9	35.0	35.0	34.1	33.4	32.3	32.6	31.8	30.8	29.4
	2%	WB	23.2	24.4	26.0	27.7	28.0	27.7	27.5	27.4	27.4	26.8	25.6	24.1
		MCDB	28.2	31.2	33.9	35.0	34.0	33.0	32.2	31.4	31.7	31.3	30.3	28.4
	5%	WB	22.5	23.6	25.2	27.1	27.4	27.2	27.0	26.9	27.0	26.3	25.1	23.3
		MCDB	27.3	30.5	32.9	34.4	33.3	32.2	31.4	30.8	31.2	30.7	29.5	27.5
	10%	WB	21.7	22.9	24.4	26.4	26.9	26.8	26.6	26.5	26.5	25.8	24.5	22.5
		MCDB	26.5	29.7	32.1	33.7	32.6	31.4	30.5	30.0	30.4	29.9	28.6	26.6
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	15.9	15.3	12.9	9.4	7.2	6.3	6.1	7.2	8.7	10.7	12.3
		MCDBR	15.0	17.7	17.5	15.0	11.8	9.3	8.2	7.8	8.5	9.6	11.3	12.9
		MCWBR	7.0	7.4	6.3	4.8	3.9	3.1	3.1	3.1	3.2	3.6	4.6	5.9
	5% WB	MCDBR	12.6	15.5	15.1	13.4	10.4	8.3	7.6	7.1	8.0	8.8	9.9	11.0
		MCWBR	6.4	7.2	6.2	5.4	4.1	3.2	3.1	3.1	3.3	3.6	4.5	5.3
(40) Clear-Sky Solar Irradiance	taub		0.448	0.555	0.727	0.692	0.506	0.452	0.448	0.468	0.494	0.534	0.434	0.432
	taud		2.170	1.886	1.543	1.664	2.169	2.349	2.352	2.277	2.185	2.032	2.278	2.239
	Ebn at Noon		824	752	644	675	805	841	846	834	811	761	833	826
	Edh at Noon		140	196	286	255	153	126	126	136	148	167	125	128
(44) All-Sky Solar Radiation	RadAvg		4.42	5.13	5.31	5.77	5.26	4.70	4.32	4.29	4.55	4.55	4.57	4.19
	RadStd		0.26	0.27	0.31	0.39	0.48	0.36	0.37	0.28	0.20	0.21	0.29	0.26

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+152	+150

Nomenclature: See separate page